

SPG-BANDIT: SKILL PROFILE GUIDED BANDIT FOR TASK SELECTION IN LLM AGENT SKILL EVOLUTION

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1 INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Large language model (LLM) agents have demonstrated competence on complex tasks in diverse environments, including embodied control, web navigation, and software engineering. An important mechanism underlying this breadth of capability is *skill evolution*, through which an agent reflects on execution trajectories, extracts reusable knowledge, and accumulates that knowledge in a skill library for subsequent episodes.

Existing work has primarily addressed two components of this process. The first is *skill quality*, namely the extraction of informative and reusable knowledge from trajectories. For example, SkillRL uses reflection to identify reusable strategies from successful episodes and recurrent failure modes from unsuccessful episodes. The second is *skill utilization*, namely the organization, retrieval, and incorporation of skills into an agent’s inference-time prompt. Representative approaches include structured skill banks with category-specific retrieval, vector-database retrieval, and dynamic prompting.

A critical dimension remains comparatively underexplored: **the order in which tasks are selected for practice**. Existing systems commonly cycle uniformly through a fixed task pool, sample tasks at random, or rely on hand-designed curricula that do not adapt to an agent’s evolving competence. Since tasks can make substantially different contributions to skill progress, an adaptive selection policy may use the interaction budget more efficiently. This motivates the following question: *Can task selection conditioned on an agent’s current skill profile improve the efficiency and effectiveness of skill evolution in LLM agents?*

1.2 CONTRIBUTIONS

We propose **SPG-Bandit** (Skill Profile Guided Bandit), a framework that casts task selection as a contextual bandit problem informed by an agent’s multidimensional skill profile.

Our central idea is to use Multidimensional Item Response Theory (MIRT) to estimate latent proficiency across K skill dimensions and to select tasks that maximize expected progress on dimensions with lower estimated proficiency.

Specifically, our contributions are threefold:

1. **Problem formulation.** We formalize task selection during LLM agent skill evolution as a non-stationary contextual bandit problem. The formulation captures both task-specific differences in expected skill progress and the multidimensional nature of agent competence.
2. **Method.** We propose SPG-Bandit, which combines MIRT-based latent skill profiling with a gap-weighted sliding-window UCB policy. The method maintains an online proficiency estimate and selects tasks with high predicted progress on underdeveloped dimensions while accounting for estimation uncertainty.
3. **General framework.** We provide a modular framework that can be integrated with existing skill-evolution methods and supports systematic evaluation of task-selection efficiency and downstream task performance.

2 PROBLEM FORMULATION

2.1 NOTATION AND SETUP

Consider an LLM agent interacting with a fixed task pool $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_M\}$ containing M tasks. At each round $t = 1, 2, \dots$, the agent selects and executes a task $\tau_t \in \mathcal{T}$ and observes a binary outcome $y_t \in \{0, 1\}$, indicating success or failure given its current skills.

The agent has a *latent skill profile* $\mathbf{s}_t \in [0, 1]^K$, which represents proficiency across K distinct skill dimensions. This profile is unobserved and inferred from task outcomes. We define the estimated *skill progress* after executing τ_t as $\delta_t = \mathbf{s}_{t+1} - \mathbf{s}_t$. Accordingly, δ_t represents the change in the filtered latent-profile estimate following the observed task outcome and reflection update.

Each task τ is associated with a *task embedding* $\mathbf{e}_\tau \in \mathbb{R}^{d_e}$, where $d_e \in \mathbb{N}_+$, typically obtained by applying a sentence encoder to its natural-language goal description.

2.2 PROBLEM DEFINITION

We formalize adaptive task selection as a *non-stationary contextual bandit* problem. At each round t :

1. The agent observes its context: its current skill profile $\mathbf{s}_t$ and the embedding $\mathbf{e}_\tau$ of every task $\tau \in \mathcal{T}$.
2. The agent selects an arm $\tau_t \in \mathcal{T}$.
3. The agent executes τ_t , observes the binary outcome y_t , and updates its skills through reflection.
4. The latent skill profile updates to $\mathbf{s}_{t+1}$.

We define a *gap weight* vector:

$$\mathbf{g}_t = \text{softmax}\left(\frac{\mathbf{1} - \mathbf{s}_t}{\gamma_g}\right), \quad (1)$$

where $\gamma_g > 0$ is a temperature parameter. We use gap-weighted skill progress as the immediate bandit reward:

$$r_t = \mathbf{g}_t^\top \delta_t. \quad (2)$$

The goal is to maximize the expected cumulative reward:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T r_t \mid \tau_t \sim \pi(\cdot \mid \mathbf{s}_t, \{\mathbf{e}_\tau\}) \right], \quad (3)$$

where π denotes the task-selection policy and T is the number of rounds in the bandit phase. This reward prioritizes progress on underdeveloped dimensions without requiring an auxiliary objective.

We learn a surrogate model $\hat{\delta}_t(\tau)$ and use $\mathbf{g}_t^\top \hat{\delta}_t(\tau)$ as the bandit’s estimated reward.

3 METHOD

3.1 OVERVIEW

SPG-Bandit consists of a warmup phase and a bandit phase. During **warmup**, the agent samples tasks uniformly to obtain an initial set of interaction outcomes. A sequential MIRT model with temporal prior $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)$ then infers a latent-profile sequence, corresponding skill deltas $\{\delta_t\}$, and task item parameters. These estimates provide supervision for an MLP that maps a task embedding and the current profile to a representation suitable for predicting skill progress.

During the **bandit phase**, the method computes a skill-gap vector $\mathbf{g}_t$ from the current profile and evaluates every candidate task using a sliding-window neural linear UCB score. The selected task maximizes predicted gap-weighted progress together with an uncertainty bonus. The sliding window discards outdated interactions so that the ridge estimator remains responsive to changes in the context-to-progress relationship. After execution, an online Bayesian MIRT update produces the next profile estimate, and the resulting observation is inserted into the window.

3.2 KEY COMPONENTS

Multidimensional IRT (MIRT). We associate each task τ with a K -dimensional discrimination vector $\mathbf{a}_\tau \in \mathbb{R}^K$ and a scalar difficulty $d_\tau \in \mathbb{R}$. Given the agent’s current skill profile $\mathbf{s} \in [0, 1]^K$, its success probability on task τ is:

$$p_\tau(\mathbf{s}) = \sigma(\mathbf{a}_\tau^\top \mathbf{s} - d_\tau), \quad (4)$$

where $\sigma(x) = 1/(1 + e^{-x})$. We impose the element-wise constraint $\mathbf{a}_\tau \geq 0$ and the scalar constraint $d_\tau \geq 0$. The first constraint ensures that increasing any component of $\mathbf{s}$ cannot reduce the predicted success probability. Consequently, the model aligns greater proficiency with stronger task performance. The nonnegative difficulty constraint further encodes a conservative prior over task difficulty and makes $\mathbf{s}$ a directionally interpretable measure of agent capability.

After collecting N warmup outcomes $\{(\tau_t, y_t)\}_{t=1}^N$, we estimate skill profiles and item parameters by alternating optimization. The temporal prior $p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)$ regularizes adjacent profiles and stabilizes inference. As in standard IRT, reliable estimation additionally requires a fixed scale convention and sufficient task coverage.

E-step. With item parameters $\{(\mathbf{a}_\tau, d_\tau)\}$ fixed, infer $\mathbf{s}_{1:N+1}$ by maximizing the joint log-posterior

$$\log p(\mathbf{s}_1) + \sum_{t=1}^N [\log p(y_t \mid \mathbf{s}_t, \mathbf{a}_{\tau_t}, d_{\tau_t}) + \log p(\mathbf{s}_{t+1} \mid \mathbf{s}_t)],$$

where $p(\mathbf{s}_1)$ is a truncated Gaussian prior on $[0, 1]^K$ centered at 0.5. In practice, the temporal prior is also a truncated Gaussian on $[0, 1]^K$, centered at $\mathbf{s}_t$.

M-step. With profiles $\{\mathbf{s}_t\}$ fixed, maximize each task’s observation log-likelihood with respect to $(\mathbf{a}_\tau, d_\tau)$ using L-BFGS-B, subject to $\mathbf{a}_\tau \geq 0$, $d_\tau \geq 0$, and ℓ_2 regularization on $\mathbf{a}_\tau$. Under these constraints, $\mathbf{s} \uparrow \Rightarrow p_\tau(\mathbf{s}) \uparrow$, so greater proficiency monotonically implies a higher predicted success probability.

We alternate the E-step and M-step until the objective converges.

Online update. During the bandit phase, each observation (y_t, τ_t) refines the profile without rerunning the full alternating procedure. Given $\mathbf{s}_t$ and $(\mathbf{a}_{\tau_t}, d_{\tau_t})$, the updated profile is obtained as:

$$\mathbf{s}_{t+1} = \arg \max_{\mathbf{s} \in [0, 1]^K} \log p(y_t \mid \mathbf{s}, \mathbf{a}_{\tau_t}, d_{\tau_t}) + \log p(\mathbf{s} \mid \mathbf{s}_t). \quad (5)$$

Delta prediction via MLP. To predict the improvement associated with a candidate task, we train a two-layer MLP Φ that maps the concatenation of task embedding $\mathbf{e}_\tau$ and current profile $\mathbf{s}$ to a feature vector $\phi \in \mathbb{R}^{d_f}$, where $d_f \in \mathbb{N}_+$. For hidden width d_h , the parameters satisfy $\mathbf{W}_1 \in \mathbb{R}^{d_h \times (d_e + K)}$, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{d_f \times d_h}$, and $\mathbf{b}_2 \in \mathbb{R}^{d_f}$:

$$\mathbf{h} = \max(\mathbf{W}_1[\mathbf{e}_\tau; \mathbf{s}] + \mathbf{b}_1, 0), \quad (6)$$

$$\phi = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2. \quad (7)$$

A linear pretraining head $\mathbf{W}_3 \in \mathbb{R}^{d_f \times K}$ predicts the expected delta as $\hat{\delta} = \mathbf{W}_3^\top \phi$. The MLP is trained by minimizing mean squared error on the warmup dataset $\{([\mathbf{e}_{\tau_i}; \mathbf{s}_i], \delta_i)\}_{i=1}^N$. After pretraining, $\mathbf{W}_3$ is discarded and Φ is fixed. Bandit decisions use the sliding-window ridge head $\widehat{\mathbf{W}}_t$.

Gap-weighted UCB selection. To accommodate a non-stationary context-to-progress relationship, we maintain a sliding window containing the ℓ most recent *completed* interactions and fit a local ridge-regression head within that window. Let $\mathcal{W}_t$ denote the indices stored in the buffer immediately before bandit round t . The buffer is initialized with the final ℓ warmup observations. For each buffered observation i , define $\phi_i = \Phi([\mathbf{e}_{\tau_i}; \mathbf{s}_i])$. The resulting covariance and response matrices are:

$$\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t, \quad \mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \quad \mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \delta_i^\top, \quad (8)$$

For a candidate task τ , let $\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}_t])$. The predicted delta is $\hat{\delta}_t(\tau) = \widehat{\mathbf{W}}_t^\top \phi_{t,\tau}$. The matrix $\mathbf{A}_t$ captures epistemic uncertainty through $\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}$. The UCB score is:

$$\text{score}_t(\tau) = \mathbf{g}_t^\top \hat{\delta}_t(\tau) + \beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}, \quad (9)$$

The first term favors tasks with high predicted gap-weighted progress. The second term is a confidence bonus for the scalarized reward $\mathbf{g}_t^\top \delta_t$, where $\beta_t > 0$ is selected according to a ridge-regression confidence bound.

Sliding-window update. After observing δ_t , we append $(\phi_{t,\tau_t}, \delta_t)$ to the window and evict its oldest entry when the window exceeds size ℓ . The incremental updates are:

$$\mathbf{A}_{t+1} = \mathbf{A}_t + \phi_{t,\tau_t} \phi_{t,\tau_t}^\top - \phi_{\text{old}} \phi_{\text{old}}^\top, \quad (10)$$

$$\mathbf{B}_{t+1} = \mathbf{B}_t + \phi_{t,\tau_t} \delta_t^\top - \phi_{\text{old}} \delta_{\text{old}}^\top, \quad (11)$$

$$\widehat{\mathbf{W}}_{t+1} = \mathbf{A}_{t+1}^{-1} \mathbf{B}_{t+1}. \quad (12)$$

The window size ℓ controls the trade-off between estimation quality, which benefits from a larger sample, and responsiveness to non-stationarity, which benefits from recent observations.

3.3 ALGORITHM

Algorithm 1 summarizes the complete procedure.

4 THEORETICAL ANALYSIS

Because the relationship between context and skill progress may change over time, expected task rewards can be non-stationary. In this section, $t \in \{1, \dots, T\}$ indexes only bandit-phase rounds. We analyze the dynamic *pseudo-regret* of SPG-Bandit under the sliding-window ridge estimator described in Section 3.

4.1 NON-STATIONARY SETTING

For each candidate task τ , define $\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}_t])$ and the conditional mean skill progress

$$\boldsymbol{\mu}_t(\tau) = \mathbb{E}[\delta_t(\tau) \mid \mathbf{s}_t].$$

The expected scalar reward is $q_t(\tau) = \mathbf{g}_t^\top \boldsymbol{\mu}_t(\tau)$. Let $\tau_t^* \in \arg \max_{\tau \in \mathcal{T}} q_t(\tau)$ be the dynamic oracle action. We evaluate the dynamic pseudo-regret

$$R_T = \sum_{t=1}^T [q_t(\tau_t^*) - q_t(\tau_t)]. \quad (13)$$

This quantity is a pseudo-regret because it compares conditional expected rewards rather than realized noisy outcomes. Let $\mathcal{F}_{t-1}$ denote the history available before round t . Boundedness of $\mathbf{s}_t$ alone does not imply sublinear non-stationarity. We therefore impose a separate variation condition on the time-varying reward model.

Algorithm 1 SPG-Bandit

Input: Task pool $\mathcal{T}$, warmup steps $N \geq \ell$, bandit horizon T , window size ℓ , feature dimension d_f , gap temperature γ_g , confidence radii $\{\beta_t\}$, regularizer $\lambda > 0$, skill dimensions K

Phase 1: Warmup

for $i = 1$ to N **do**

Select $\tau_i \in \mathcal{T}$ uniformly

Execute τ_i , observe $y_i \in \{0, 1\}$, and perform the skill-evolution update

Store (τ_i, y_i)

end for

Run sequential MIRT EM on $\{(\tau_i, y_i)\}_{i=1}^N \rightarrow$ profiles $\{\mathbf{s}_i\}_{i=1}^{N+1}$, deltas $\{\delta_i\}_{i=1}^N$, item parameters $\{\mathbf{a}_\tau, d_\tau\}_{\tau \in \mathcal{T}}$

Train and freeze MLP Φ on $\{([\mathbf{e}_{\tau_i}; \mathbf{s}_i], \delta_i)\}_{i=1}^N$

Initialize $\mathcal{W}$ with the latest ℓ warmup tuples (ϕ_i, δ_i) , where $\phi_i = \Phi([\mathbf{e}_{\tau_i}; \mathbf{s}_i])$

Set $\mathbf{A} = \lambda \mathbf{I}_{d_f} + \sum_{(\phi_i, \delta_i) \in \mathcal{W}} \phi_i \phi_i^\top$, $\mathbf{B} = \sum_{(\phi_i, \delta_i) \in \mathcal{W}} \phi_i \delta_i^\top$, $\widehat{\mathbf{W}} = \mathbf{A}^{-1} \mathbf{B}$, and $\mathbf{s} = \mathbf{s}_{N+1}$

Phase 2: Bandit

for $t = 1$ to T **do**

Compute gap $\mathbf{g}_t = \text{softmax}((1 - \mathbf{s})/\gamma_g)$

for each $\tau \in \mathcal{T}$ **do**

$\phi_{t,\tau} = \Phi([\mathbf{e}_\tau; \mathbf{s}])$, $\hat{\delta}_t(\tau) = \widehat{\mathbf{W}}^\top \phi_{t,\tau}$

$\text{score}_t(\tau) = \mathbf{g}_t^\top \hat{\delta}_t(\tau) + \beta_t \|\mathbf{g}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}^{-1} \phi_{t,\tau}}$

end for

$\tau_t = \arg \max_{\tau \in \mathcal{T}} \text{score}_t(\tau)$

Execute τ_t and observe y_t

$\mathbf{s}_{\text{new}} \leftarrow \text{BAYESUPDATE}(\mathbf{s}, \mathbf{a}_{\tau_t}, d_{\tau_t}, y_t)$

$\delta_t \leftarrow \mathbf{s}_{\text{new}} - \mathbf{s}$ and $\mathbf{s} \leftarrow \mathbf{s}_{\text{new}}$

Append $(\phi_{t,\tau_t}, \delta_t)$ to $\mathcal{W}$

$\mathbf{A} += \phi_{t,\tau_t} \phi_{t,\tau_t}^\top$ and $\mathbf{B} += \phi_{t,\tau_t} \delta_t^\top$

if $|\mathcal{W}| > \ell$ **then**

Pop oldest $(\phi_{\text{old}}, \delta_{\text{old}})$ from $\mathcal{W}$

$\mathbf{A} -= \phi_{\text{old}} \phi_{\text{old}}^\top$ and $\mathbf{B} -= \phi_{\text{old}} \delta_{\text{old}}^\top$

end if

$\widehat{\mathbf{W}} = \mathbf{A}^{-1} \mathbf{B}$

end for

4.2 ASSUMPTIONS

1. **(Time-varying linear delta model.)** There are matrices $\{\mathbf{W}_t^*\}_{t=1}^T$, where $\mathbf{W}_t^* \in \mathbb{R}^{d_f \times K}$, such that $\delta_t(\tau) = \mathbf{W}_t^{*\top} \phi_{t,\tau} + \varepsilon_t(\tau)$, with $\mathbb{E}[\varepsilon_t(\tau) \mid \mathcal{F}_{t-1}, \tau] = \mathbf{0}$.
2. **(Boundedness and noise.)** $\|\phi_{t,\tau}\|_2 \leq 1$, $\|\mathbf{W}_t^*\|_F \leq S$, and, for every $\mathbf{u}$ satisfying $\|\mathbf{u}\|_2 \leq 1$, the scalar $\mathbf{u}^\top \varepsilon_t(\tau)$ is conditionally σ -sub-Gaussian.
3. **(Parameter-path variation.)** The reward-model variation is $P_T = \sum_{t=1}^{T-1} \|\mathbf{W}_{t+1}^* - \mathbf{W}_t^*\|_F$.
4. **(Calibrated window and coverage.)** Before the first bandit round, warmup provides ℓ calibration observations generated under $\mathbf{W}_1^*$. At each bandit round, the window contains exactly ℓ observations and satisfies $\sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top \succeq \kappa \ell \mathbf{I}_{d_f}$ for a constant $\kappa > 0$.

Assumption 1 matches the algorithmic regression target, which is δ_t rather than $\mathbf{s}_{t+1}$. Assumption 2 controls the scale of features, parameters, and noise. Assumption 3 quantifies non-stationarity through the parameter path. Assumption 4 isolates the effect of non-stationarity from an ill-conditioned design matrix. For a failure probability $\rho \in (0, 1)$, Appendix A provides a detailed high-probability proof.

4.3 DYNAMIC REGRET DECOMPOSITION

Let $w_t(\tau) = \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}$ and $\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t$. Under Assumptions 1–4, Appendix A proves that, with probability at least $1 - \rho$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 w_t(\tau) + D_t,$$

where $\beta_t = \tilde{O}(\sigma \sqrt{K d_f} + \sqrt{\lambda} S)$ is a confidence radius and D_t is the explicitly bounded bias from stale observations. Since the policy selects the task with the largest UCB score, approximate optimism yields

$$q_t(\tau_t^*) - q_t(\tau_t) \leq 2\beta_t \|\mathbf{g}_t\|_2 w_t(\tau_t) + 2D_t.$$

Because $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Assumption 4 gives $w_t(\tau) \leq (\lambda + \kappa \ell)^{-1/2}$ for every candidate task. The detailed counting argument in Appendix A gives $\sum_{t=1}^T D_t \leq \ell^2 P_T / \sqrt{\lambda(\lambda + \kappa \ell)}$. Therefore, with $\beta = \max_{t \leq T} \beta_t$,

$$R_T \leq \underbrace{\frac{2\beta T}{\sqrt{\lambda + \kappa \ell}}}_{\text{estimation}} + \underbrace{\frac{2\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa \ell)}}}_{\text{drift}}. \quad (14)$$

4.4 OPTIMAL WINDOW AND REGRET BOUND

For $P_T > 0$, when $\kappa \ell \geq \lambda$, balancing the two terms yields the following feasible integer window size:

$$\ell^* = \min \left\{ T, \max \left\{ \left\lceil \frac{\lambda}{\kappa} \right\rceil, \left\lceil \left(\frac{\beta T \sqrt{\lambda}}{P_T} \right)^{1/2} \right\rceil \right\} \right\}, \quad (15)$$

and the corresponding cumulative regret bound, up to universal constants:

$$R_T = O \left(\frac{(\beta T)^{3/4} P_T^{1/4}}{\kappa^{1/2} \lambda^{1/8}} \right). \quad (16)$$

If $P_T = 0$, the drift term vanishes and $\ell = T$ minimizes the displayed bound. If $P_T = o(T)$ and β , κ^{-1} , and λ^{-1} grow at most polylogarithmically, then the bound is sublinear. For example, if $P_T = O(T^\nu)$ for $\nu < 1$, then

$$R_T = \tilde{O} \left(T^{(3+\nu)/4} \right) = o(T). \quad (17)$$

A DETAILED DYNAMIC PSEUDO-REGRET ANALYSIS

This appendix proves the dynamic pseudo-regret bound in Section 4. Every bound below is conditional on Assumptions 1–4. In particular, the coverage assumption is essential because the stated result does not hold for an arbitrary sequence of feature vectors.

A.1 INDEXING, FILTRATION, AND NOTATION

The bandit phase is indexed by $t \in \{1, \dots, T\}$. To avoid a boundary case in the first ℓ rounds, we regard the ℓ calibration observations from warmup as rounds $1 - \ell, \dots, 0$. By Assumption 4, they are generated with parameter $\mathbf{W}_1^*$. For every bandit round t , define

$$\mathcal{W}_t = \{t - \ell, \dots, t - 1\}. \quad (18)$$

Thus, $|\mathcal{W}_t| = \ell$ for every t . This indexing is equivalent to the buffer implementation in Algorithm 1.

Let $\mathcal{F}_{t-1}$ be the sigma-field generated by the calibration observations and all selected tasks and observed outcomes before bandit round t . For an observed task at index i , abbreviate $\phi_i = \phi_{i,\tau_i}$. Assumption 1 gives

$$\delta_i = \mathbf{W}_i^* \phi_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid \mathcal{F}_{i-1}, \tau_i] = \mathbf{0}. \quad (19)$$

For calibration indices $i \leq 0$, set $\mathbf{W}_i^* = \mathbf{W}_1^*$. Consequently, calibration data do not contribute to the parameter-path variation.

At the beginning of round t , ridge regression uses

$$\begin{aligned} \mathbf{A}_t &= \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \\ \mathbf{B}_t &= \sum_{i \in \mathcal{W}_t} \phi_i \delta_i^\top, \\ \widehat{\mathbf{W}}_t &= \mathbf{A}_t^{-1} \mathbf{B}_t. \end{aligned} \quad (20)$$

For a candidate task τ , write

$$w_t(\tau) = \|\phi_{t,\tau}\|_{\mathbf{A}_t^{-1}} = \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}. \quad (21)$$

A.2 A UNIFORM RIDGE-REGRESSION CONFIDENCE BOUND

For a failure probability $\rho \in (0, 1)$, define

$$\beta_t = \sigma \sqrt{2K \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)} + \sqrt{\lambda} S. \quad (22)$$

Let

$$\mathbf{Z}_t = \sum_{i \in \mathcal{W}_t} \phi_i \varepsilon_i^\top. \quad (23)$$

For the k -th output coordinate and a fixed window $\mathcal{W}_t$, the self-normalized martingale inequality gives

$$\left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \mathbf{e}_k \right\|_2 \leq \sigma \sqrt{2 \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)}. \quad (24)$$

Applying a union bound over the K output coordinates and the T windows makes the inequality hold simultaneously for every $k \in \{1, \dots, K\}$ and $t \leq T$, with probability at least $1 - \rho$. The factor KT inside the logarithm records this union bound. Summing the squared coordinate bounds yields

$$\left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \right\|_F \leq \sigma \sqrt{2K \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)}. \quad (25)$$

A.3 PREDICTION ERROR DECOMPOSITION

Substituting equation 19 into equation 20 gives

$$\mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top \mathbf{W}_i^* + \mathbf{Z}_t. \quad (26)$$

Since $\mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top$, adding and subtracting $\mathbf{W}_t^*$ gives

$$\begin{aligned} \widehat{\mathbf{W}}_t - \mathbf{W}_t^* &= \mathbf{A}_t^{-1} \mathbf{Z}_t - \lambda \mathbf{A}_t^{-1} \mathbf{W}_t^* \\ &\quad + \mathbf{A}_t^{-1} \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top (\mathbf{W}_i^* - \mathbf{W}_t^*). \end{aligned} \quad (27)$$

The three terms are the stochastic estimation term, the regularization term, and the stale-window bias.

Fix a candidate task τ . Cauchy–Schwarz and equation 25 give

$$|\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \mathbf{Z}_t \mathbf{g}_t| \leq w_t(\tau) \left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \right\|_F \|\mathbf{g}_t\|_2. \quad (28)$$

Moreover, since $\mathbf{A}_t \succeq \lambda \mathbf{I}_{d_f}$,

$$|\lambda \phi_{t,\tau}^\top \mathbf{A}_t^{-1} \mathbf{W}_t^* \mathbf{g}_t| \leq \sqrt{\lambda} S w_t(\tau) \|\mathbf{g}_t\|_2. \quad (29)$$

For the stale-window bias, define

$$V_t^{\text{win}} = \sum_{i \in \mathcal{W}_t} \|\mathbf{W}_i^* - \mathbf{W}_t^*\|_F. \quad (30)$$

For every $i \in \mathcal{W}_t$, Cauchy–Schwarz and $\mathbf{A}_t \succeq \lambda \mathbf{I}_{d_f}$ imply

$$|\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_i| \leq \left\| \mathbf{A}_t^{-1/2} \phi_{t,\tau} \right\|_2 \left\| \mathbf{A}_t^{-1/2} \phi_i \right\|_2 \quad (31)$$

$$\leq \frac{w_t(\tau)}{\sqrt{\lambda}}. \quad (32)$$

Using $\|\phi_i\|_2 \leq 1$, the bias term in equation 27 is bounded by

$$\frac{w_t(\tau) \|\mathbf{g}_t\|_2}{\sqrt{\lambda}} V_t^{\text{win}}. \quad (33)$$

Assumption 4 gives $\mathbf{A}_t \succeq (\lambda + \kappa\ell) \mathbf{I}_{d_f}$, and therefore

$$w_t(\tau) \leq \frac{1}{\sqrt{\lambda + \kappa\ell}}. \quad (34)$$

Define

$$D_t = \frac{V_t^{\text{win}}}{\sqrt{\lambda(\lambda + \kappa\ell)}}. \quad (35)$$

Combining equation 27–equation 35 proves that, simultaneously for all $t \leq T$ and $\tau \in \mathcal{T}$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 w_t(\tau) + D_t. \quad (36)$$

A.4 SUMMING THE DRIFT ALLOWANCES

Let $v_j = \|\mathbf{W}_{j+1}^* - \mathbf{W}_j^*\|_F$ for $j \geq 1$, and set $v_j = 0$ for $j \leq 0$. The triangle inequality gives, for $i < t$,

$$\|\mathbf{W}_i^* - \mathbf{W}_t^*\|_F \leq \sum_{j=i}^{t-1} v_j. \quad (37)$$

Thus,

$$\sum_{t=1}^T V_t^{\text{win}} \leq \sum_{t=1}^T \sum_{i=t-\ell}^{t-1} \sum_{j=i}^{t-1} v_j \quad (38)$$

$$\leq \ell^2 \sum_{j=1}^{T-1} v_j = \ell^2 P_T. \quad (39)$$

For each fixed increment v_j , there are at most ℓ windows that contain it and at most ℓ indices within each such window. Combining equation 35 and equation 39 gives

$$\sum_{t=1}^T D_t \leq \frac{\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa\ell)}}. \quad (40)$$

A.5 FROM CONFIDENCE TO DYNAMIC PSEUDO-REGRET

Define $U_t(\tau) = \beta_t \|\mathbf{g}_t\|_2 w_t(\tau)$. Equation 36 gives, for every candidate task,

$$q_t(\tau) \leq \mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \phi_{t,\tau} + U_t(\tau) + D_t. \quad (41)$$

The policy selects τ_t to maximize the first two terms on the right-hand side. Therefore,

$$q_t(\tau_t^*) \leq \mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \phi_{t,\tau_t} + U_t(\tau_t) + D_t \quad (42)$$

$$\leq q_t(\tau_t) + 2U_t(\tau_t) + 2D_t. \quad (43)$$

The second inequality follows by applying the lower counterpart of equation 36 to the selected task.

Since $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Equation 34 gives

$$\sum_{t=1}^T U_t(\tau_t) \leq \frac{\beta T}{\sqrt{\lambda + \kappa \ell}}, \quad \beta = \max_{t \leq T} \beta_t. \quad (44)$$

Summing equation 43, then applying equation 40 and equation 44, establishes

$$R_T \leq \frac{2\beta T}{\sqrt{\lambda + \kappa \ell}} + \frac{2\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa \ell)}}. \quad (45)$$

When $\kappa \ell \geq \lambda$, the two terms in equation 45 have orders $\beta T/(\sqrt{\kappa \ell})$ and $\ell^{3/2} P_T/(\sqrt{\kappa \lambda})$. For $P_T > 0$, equating these orders gives $\ell = (\beta T \sqrt{\lambda}/P_T)^{1/2}$. Rounding upward, enforcing $\kappa \ell \geq \lambda$, and truncating at T yields the window size in Section 4. If $P_T = 0$, the drift term is zero and the largest feasible window is optimal. $\square$